Objective function (needs to be minimized) = $\underbrace{\left[\sum_{i=1}^{n} \ell(y_i, f_{m-1}(x_i) + \gamma_{im}\right]}_{\text{Loss function}} + \underbrace{0.5 v \sum_{j=1}^{T} \gamma_{jm}^2}_{\text{squared penalty}} + aT$ → allows pruning

$(x-a)$ = Distance from expansion

Taylor Series → $f(x) \approx f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2$ $\quad$ i = current index

Assume $f(a) = \ell(y, f_{m-1}(x_i))$

$\frac{f'(a)}{1!}(x-a) \rightarrow g_i \cdot \gamma_m$ $\quad$ $f_{m-1}(x_i)$ is 'a', the current location moving by γ_m

$\frac{f''(a)}{2!}(x-a)^2 \rightarrow 0.5 h_i \gamma_m^2$

Hence $\approx \ell(y, f_{m-1}(x_i)) + \gamma_{im} g_i + 0.5 h_i \gamma^2$

Substitute back into full objective formula =

$$\sum_{i=1}^{n} \ell(y, f_{m-1}(x_i)) + \sum_{i=1}^{n} \gamma_{im} g_i + 0.5 \sum_{i=1}^{n} h_i \gamma_{im}^2 + 0.5 v \sum_{j=1}^{|T|} \gamma_{jm}^2 + aT$$

Re-arrange by enumerating over all leaves first, since all observations in the same leaf gets γ_{jm}

$$= \sum_{i=1}^{n} \ell(y_i, f_{m-1}(x_i)) + \sum_{j=1}^{T} \left[\sum_{x \in R_{jm}} \gamma_{jm} g_i + 0.5 \left(\sum_{x_i \in R_{jm}} h_i + v\right) \gamma_{jm}^2\right] + aT$$

1st Derivative with $\gamma_{jm} = 0$,

$$= \underbrace{\cancel{\sum_{i=1}^{n} \ell(y_i, f_{m-1}(x_i))}}_{\text{Constant}} + \sum_{j=1}^{T} \left[\sum_{x \in R_{jm}} \gamma_{jm} g_i + 0.5 \left(\sum_{x_i \in R_{jm}} h_i + v\right) \gamma_{jm}^2\right] + \underbrace{\cancel{aT}}_{\text{constant}}$$

Differentiate γ_{jm} on $\sum_{x \in R_{jm}} \gamma_{jm} g_i + 0.5 \left(\sum_{x_i \in R_{jm}} h_i + v\right) \gamma_{jm}^2$

$$0 = \sum_{x \in R_{jm}} g_i + \left(\sum h_i + v\right) \gamma_{jm}$$

$$\gamma_{jm} = \frac{\sum_{x \in R_{jm}} g_i}{\sum_{x_i \in R_{jm}} h_i + v}$$

MART predicts the mean residual in each leaf. XGBoost does the same with squared loss (with v regularization)

Squared loss = $(y_i - f_{m-1}(x_i))^2$

Arbitrarily added for convenience

$$\ell = 0.5 (y_i - f_{m-1}(x_i))^2$$

$$\frac{\partial}{\partial f_{m-1}(x_i)} = 0.5 \cdot 2 (y_i - f_{m-1}(x_i))(-1)$$

First Derivative

$$g_i = -(y_i - f_{m-1}(x_i))$$

Second Derivative

$$h_i = 1$$ → multiply −1 to switch minimization to maximization

Hence, $\gamma_{jm} = \frac{\sum_{x_i \in R} (y_i - f_{m-1}(x_i))}{n_{jm} + v} = \frac{\text{sum of residuals}}{\text{sum of residuals} + v}$